

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Linear Spaces

Linear Algebra

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Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

A **groupoid** $(E, \star)$ is a set E equipped with a binary law of composition $\star$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

A **semigroup** is a groupoid whose law of composition is associative. It is said to have an **identity element** if there exists an element $e \in S$ such that

$$\star(e, a) = \star(a, e) = a.$$

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

A **group** is a set G together with an associative binary law of composition $\star : G \times G \rightarrow G$ called **multiplication** and having the following properties:

- i) There exists a unique element $e \in G$ called the **identity** such that $e \star g = g \star e = g$ for all $g \in G$.
- ii) For every element $g \in G$, there exists an element g^{-1} , called the **inverse** of g , such that $g \star g^{-1} = g^{-1} \star g = e$.

To emphasize the binary operation of a group, we designate it as $(G, \star)$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Remark

*If the underlying set has a finite number of elements, the group is said to be **finite**, and its number of elements, denoted by $|G|$, is called the **order** of G . We can also have an infinite group whose cardinality can be countable or continuous.*

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Remark

Given an element $a \in G$, we write

$$a^k \equiv \underbrace{a \star a \star \cdots \star a}_{k \text{ times}}, \quad a^{-m} \equiv \underbrace{a^{-1} \star a^{-1} \star \cdots \star a^{-1}}_{m \text{ times}},$$

and note that

$$a^i \star a^j = a^{i+j} \quad \text{for all } i, j \in \mathbb{Z}.$$

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

A group $(G, \star)$ is said to be **Abelian** or **commutative** if $\star$ is commutative. It is customary to denote the binary operation of an Abelian group by $+$.



Niels Henrik Abel
1802-1829

Historical Notes

Niels Henrik Abel was the second of seven children, son of a Lutheran minister with a small parish of Norwegian coastal islands. In school he received only average marks at first, but then his mathematics teacher was replaced by a man only seven years older than Abel. Abel's alcoholic father died in 1820, leaving almost no financial support for his young prodigy, who became responsible for supporting his mother and family. His teacher, **Holmboe**, recognizing his talent for mathematics, raised money from his colleagues to enable Abel to attend Christiania (modern Oslo) University. He entered the university in 1821, 10 years after the university was founded, and soon proved himself worthy of his teacher's accolades. His second paper, for example, contained the first example of a solution to an integral equation. Abel then received a two-year government travel grant and journeyed to Berlin, where he met the prominent mathematician **Crelle**, who soon launched what was to become the leading German mathematical journal of the nineteenth century, commonly called *Crelle's Journal*. From the start, Abel contributed important papers to Crelle's Journal, including a classic paper on power series, the scope of which clearly reflects his desire for stringency. His most important work, also published in that journal, was a lengthy treatment of elliptic functions in which Abel incorporated their inverse functions to show that they are a natural generalization of the trigonometric functions.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Example 1.1

The pair $(\mathbb{Z}, +)$ where $\mathbb{Z}$ is the set of integers and $+$ is Peano's sum, forms an Abelian group.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

A subset S of G is a **subgroup** of G if it is a group in its own right under the binary operation of G , i.e., if it contains the inverse of all its elements as well as the product of any pair of its elements.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

Let $(G, \star), (H, \star)$ be groups. A **group homomorphism** $\varphi : G \rightarrow H$ is a map such that

$$\varphi(a \star b) = \varphi(a) \star \varphi(b).$$

An **isomorphism** is a homomorphism that is also a bijection. Two groups are **isomorphic**, denoted by $G \cong H$, if there is an isomorphism $\phi : G \rightarrow H$. An isomorphism of a group onto itself is called an **automorphism**.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Example 1.2

Show that the group of isometries of a triangle under composition is isomorphic to the group of integer powers of the imaginary unit i under the usual complex multiplication.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Example 1.3

The pairs $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \times)$ are groups and the map $E : \mathbb{R} \rightarrow \mathbb{R}^+$ is a group homomorphism.

Definition

A **ring** R is a set with two binary laws of composition $(+, \bullet)$, called **addition** and **multiplication**, respectively, such that R is an abelian group with respect to $+$, a semigroup with respect to $\bullet$ and $\bullet$ is **distributive** over $+$, i.e.

$$\bullet(p, +(q, r)) = +(\bullet(p, q), \bullet(p, r)) \quad \text{for } p, q, r \in R.$$

We usually write

$$+ : (a, b) \rightarrow a + b \quad \text{and} \quad \bullet : (a, b) \rightarrow ab \quad \text{for } a, b \in R.$$

Furthermore, R is said to be **commutative** [respectively, a **ring with identity**] if $(R, \bullet)$ is a commutative semigroup [respectively, a semigroup with identity].

Definition

A **body**[†] (or a **field**) $\mathbb{K}$ is a commutative ring with identity, denoted by 1, in which every element $a \neq 0$ has an inverse $a^{-1} \in \mathbb{K}$ such that $aa^{-1} = 1$.

[†]The notation $\mathbb{K}$ comes from the German word "Körper" (meaning "body")

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Example 1.4

The set $\mathbb{C}$ of all complex numbers is a body under the usual operations as is the set $\mathbb{R}$ of all real numbers.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Remark

The only bodies of interest in this course are $\mathbb{R}$ and $\mathbb{C}$ (the real and complex bodies, respectively), so for our purposes, $\mathbb{K}$ is either $\mathbb{R}$ or $\mathbb{C}$.

Definition

An Abelian group V is called a **linear space** (or **vector space**) over $\mathbb{K}$ if it is equipped with a body $\mathbb{K}$ and a map from $\mathbb{K} \times \mathcal{V}$ to $\mathcal{V}$, written $(\alpha, v) \mapsto \alpha v$ such that

$$\text{i) } \alpha(\beta v) = (\alpha\beta)v,$$

$$\text{ii) } (\alpha + \beta)v = \alpha v + \beta v,$$

$$\text{iii) } \alpha(v + w) = \alpha v + \alpha w,$$

$$\text{iv) } 1 \cdot v = v,$$

for all $v, w \in \mathcal{V}, \alpha, \beta \in \mathbb{K}$.

Definition

The elements of a linear space $\mathcal{V}$ are called **vectors** and the elements of the underlying body $\mathbb{K}$ are termed **scalars**. The map from $\mathbb{K} \times \mathcal{V}$ to $\mathcal{V}$, whose image is usually denoted by juxtaposition, is called **scalar multiplication**. The law of composition defining the Abelian group, denoted by $+$, is called **vector addition** and the identity element thereof is termed the **zero vector**. The inverse of any vector u is denoted by $-u$. A vector space over $\mathbb{C}$ (respectively, $\mathbb{R}$) is said to be a **complex** (respectively, **real**) **linear space**.

Example 1.5

(Some linear spaces)

- i) $\mathbb{R}$ is a linear space over the real body.
- ii) $\mathbb{C}$ is a linear space over the real body.
- iii) $\mathbb{C}$ is a linear space over the complex body.
- iv) The set $\mathcal{P}[z]$ of all polynomials with complex coefficients in a variable z is a linear space under the usual addition and the multiplication of a polynomial by a complex number.
- v) The set $\mathbb{C}^n$ of all complex n -tuples is a linear space under the usual definitions for addition and scalar multiplication, called the **n -dimensional complex coordinate space**.
- vi) The set $\mathbb{R}^n$ of all real n -tuples is a linear space over $\mathbb{R}$ under operations defined similarly as those of $\mathbb{C}^n$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

- vii) *The set $\mathcal{M}^{m \times n}$ of all complex matrices with m rows and n columns is a linear space under the usual addition of matrices and multiplication by complex numbers.*
- viii) *The set ℓ^2 of all sequences (z_n) such that $\sum |z_n|^2 < \infty$ is a linear space with addition and multiplication by complex numbers defined componentwise.*
- ix) *The set $\mathcal{H}(\Omega)$ of complex functions that are holomorphic on a subset $\Omega \subseteq \mathbb{C}$ of $\mathbb{C}$ is a linear space over $\mathbb{C}$.*

Einstein's Notation

Let us introduce **Einstein's summation convention**. The convention is that any lower case alphabetic index that appears exactly twice in any term of an expression is understood to be summed over all values that an index in that position can take (unless the contrary is specifically stated).

Remark

The following examples illustrate what is meant (in the case $i, j, k = 1, 2, 3$):

i) $a_i x_i$ stands for $a_1 x_1 + a_2 x_2 + a_3 x_3$,

ii) $a_{ij} b_{jk}$ stands for $a_{i1} b_{1k} + a_{i2} b_{2k} + a_{i3} b_{3k}$,

iii) $a_{ij} b_{jk} c_k = \sum_{j=1}^3 \sum_{k=1}^3 a_{ij} b_{jk} c_k$,

iv) $\frac{\partial v_i}{\partial x_i} = \frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3}$,

v) $\frac{\partial^2 \phi}{\partial x_i \partial x_i} = \frac{\partial^2 \phi}{\partial x_1^2} + \frac{\partial^2 \phi}{\partial x_2^2} + \frac{\partial^2 \phi}{\partial x_3^2}$.

Indices that are summed over are called **dummy indices** and the others **free indices**.

Definition

A collection of vectors $\{v_i\}_{i \in I}$ is said to be **finitely linearly independent** (or simply, **linearly independent**) if for every choice of finitely many distinct indices $i_1, i_2, \dots, i_N \in I$, we have

$$\alpha_j v_{i_j} = 0 \implies \alpha_j = 0 \quad \text{for all } j.$$

The collection $\{v_i\}_{i \in I}$ is said to be **linearly dependent** if it is not linearly independent. The sum $\alpha_i v_i$ is called a **finite linear combination** of $\{v_i\}_{i \in I}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

A **subspace** $\mathcal{W}$ of a linear space $\mathcal{V}$ is a nonempty subset of $\mathcal{V}$ with the property that if $u, v \in \mathcal{W}$, then $\alpha u + v$ also belongs to $\mathcal{W}$ for all $\alpha \in \mathbb{C}$.

Example 1.6

(Some linear subspaces)

- *The set $\mathcal{P}_n[z]$ of all polynomials of degree at most n with complex coefficients in a variable z is a linear subspace of $\mathcal{P}[z]$.*
- *$\mathbb{R}^m$ is a subspace of $\mathbb{R}^n$ for $m < n$.*
- *The set of all vectors along a given line (going through the origin) is a subspace of arrows in space.*

Definition

The **finite linear span** (or simply, **span**) of $\mathcal{F} = \{v_i\}_{i \in I}$ is the set of all finite linear combinations of elements of $\mathcal{F}$

$$\text{span } \mathcal{F} = \{\alpha_j v_{i_j} : N > 0, i_j \in I, \alpha_j \in \mathbb{C}\}.$$

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

A **Hamel basis** (or simply, a **basis**) of a linear space $\mathcal{V}$ is a set B of linearly independent vectors that spans all of $\mathcal{V}$. A vector space that has a finite basis is called **finite-dimensional**; otherwise, it is **infinite-dimensional**.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.7

All bases of a given finite-dimensional linear space have the same number of linearly independent vectors.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

The cardinality of a basis of a linear space $\mathcal{V}$ is called the **dimension** of $\mathcal{V}$ and denoted $\dim \mathcal{V}$. To emphasize its dependence on the scalars, $\dim_{\mathbb{C}} \mathcal{V}$ and $\dim_{\mathbb{R}} \mathcal{V}$ are also used.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

Given any vector v in an N -dimensional linear space $\mathcal{V}$ and $B = \{e_i\}$ a basis in that space, there is a unique set of scalars $\{\alpha_1, \alpha_2, \dots, \alpha_N\}$ such that $v = \alpha_i e_i$. The set $\{\alpha_i\}_{i=1}^N$ is called the **components** of v in the basis B .

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.8

The components of a vector in a given basis are unique.

Example 1.9

The following are bases for some linear spaces given in Example 1.5.

- i) The (real) number 1 is a basis for $\mathbb{R}$, which is therefore one-dimensional.
- ii) The numbers 1 and i are basis vectors for the vector space $\mathbb{C}$ over $\mathbb{R}$. Thus, this space is two-dimensional.
- iii) The number 1 is a basis for $\mathbb{C}$ over $\mathbb{C}$, and the space is one-dimensional.
- iv) A basis for $\mathcal{P}[z]$ can be formed by the monomials $1, t, t^2, \dots$. It is clear that this space is infinite-dimensional.
- v) A basis for $\mathbb{C}^n$ is given by $e_1, e_2, \dots, e_n$, where e_j is an n -tuple that has a 1 in the j th position and zeros everywhere else. This basis is called the **canonical basis** of $\mathbb{C}^n$. Clearly, the space has n dimensions.

- vii) A basis of $\mathcal{M}^{m \times n}$ is given by $e_{11}, e_{12}, \dots, e_{ij}, \dots, e_{mn}$, where e_{ij} is the $m \times n$ matrix with zeros everywhere except at the intersection of the i th row and j th column, where it has a one.
- ix) If we assume that Ω is a domain containing 0 and that f is holomorphic on Ω , then we have that the set of monomials $1, z, z^2, \dots$ forms a basis for $\mathcal{H}(\Omega)$ because by Taylor's Theorem (see Theorem 8.55 on the Complex Analysis course) we have that any function in $\mathcal{H}$ can be expanded in a convergent series around $z = 0$. Thus, this space is infinite-dimensional.

Definition

A **linear space homomorphism** (or **linear application**) from a complex linear space $\mathcal{V}$ to another space $\mathcal{W}$ is a mapping $\mathbf{T}: \mathcal{V} \rightarrow \mathcal{W}$ such that

$$\mathbf{T}(\alpha u + \beta v) = \alpha \mathbf{T}(u) + \beta \mathbf{T}(v),$$

for all $u, v \in \mathcal{V}$ and $\alpha, \beta \in \mathbb{C}$. The space of all linear space homomorphisms from $\mathcal{V}$ to $\mathcal{W}$ is denoted by $\text{Hom}(\mathcal{V}, \mathcal{W})$. A homomorphism $\mathbf{T}$ from a linear space $\mathcal{V}$ into itself is called an **endomorphism** of $\mathcal{V}$ or an **operator** on $\mathcal{V}$.

The space of endomorphisms of $\mathcal{V}$ is denoted by $\text{End}(\mathcal{V})$ rather than $\text{Hom}(\mathcal{V}, \mathcal{V})$. The action of a linear transformation on a vector is written without parentheses: $\mathbf{T}(v) = \mathbf{T}v$

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Remark

Note that the definition demands that both vector spaces have the same set of scalars: The same scalars must multiply vectors in $\mathcal{V}$ on the LHS and those in $\mathcal{W}$ on the RHS.

Definition

The **zero transformation** $\mathbf{0}$ is defined to take every vector in $\mathcal{V}$ to the zero vector of $\mathcal{W}$, in particular, the **zero operator** is defined to take every vector in $\mathcal{V}$ to the zero vector of $\mathcal{V}$. The sum of two linear transformations $\mathbf{T}$ and $\mathbf{U}$ is the linear transformation $\mathbf{T} + \mathbf{U}$, whose action on a vector $v \in \mathcal{V}$ is defined to be $(\mathbf{T} + \mathbf{U})v \equiv \mathbf{T}v + \mathbf{U}v$. Similarly, define $\alpha\mathbf{T}$ by $(\alpha\mathbf{T})v \equiv \alpha(\mathbf{T}v) = \alpha\mathbf{T}v$. Furthermore, we say that two linear applications $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{W}$ and $\mathbf{U} : \mathcal{V} \rightarrow \mathcal{W}$ are equal if and only if $\mathbf{T}(e_i) = \mathbf{U}(e_i)$ for all e_i in some basis of $\mathcal{V}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.10

$\text{Hom}(\mathcal{V}, \mathcal{W})$ is a linear space in the sense of the operations defined previously.

Definition

An **inner product** on $\mathcal{V}$ is a map $g : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$ satisfying:

- i) **(Sesquisymmetry)** $g(u, v) = g(v, u)^*$.
- ii) **(Sesquilinearity)** $g(u, \alpha v + \beta w) = \alpha g(u, v) + \beta g(u, w)$ for all $\alpha, \beta \in \mathbb{C}$.
- iii) **(Positivity)** $g(u, u) \geq 0$.
- iv) **(Definiteness)** $g(u, u) = 0$ if and only if $u = 0$.

A positive definite real inner product is also called a **Euclidean** inner product, otherwise it is called **pseudo-Euclidean**. An inner product that fails to satisfy the definiteness property is said to be **degenerate**.

A linear space $\mathcal{V}$ equipped with an inner product is called an **inner product space** (or a **pre-Hilbert space**). To emphasize the inner product of an inner product space, we sometimes designate it as $(\mathcal{V}, g)$.



Euclid of Alexandria
~325 B.C.-~265 B.C.

Historical Notes

Euclid of Alexandria is the most prominent mathematician of antiquity best known for his treatise on mathematics *The Elements*. The long lasting nature of *The Elements* must make Euclid the leading mathematics teacher of all time. However little is known of Euclid's life except that he taught at Alexandria in Egypt. *The Elements* begins with definitions and five postulates. The first three postulates are postulates of construction, for example the first postulate states that it is possible to draw a straight line between any two points. The fourth and fifth postulates are of a different nature. Postulate four states that all right angles are equal. This may seem "obvious" but it actually assumes that space is homogeneous by this we mean that a figure will be independent of the position in space in which it is placed. The famous fifth, or parallel, postulate states that one and only one line can be drawn through a point parallel to a given line. Euclid's decision to make this a postulate led to Euclidean geometry. It was not until the 19th century that this postulate was dropped and non-euclidean geometries were studied.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.11 (Properties of Inner Products)

Let $(\mathcal{V}, g)$ be an inner product space. Then

v) $g(0, v) = g(v, 0) = 0.$

vi) $g(\alpha u + \beta v, w) = \alpha^* g(u, w) + \beta^* g(v, w).$

Definition

Let u, v be vectors in an inner product space $(\mathcal{V}, g)$. Then we say that u and v are **orthogonal** if $g(u, v) = 0$. A **normal** (or **normalised**) vector e is one for which $g(e, e) = 1$. A basis $B = \{e_i\}_{i=1}^N$ in an N -dimensional linear space $\mathcal{V}$ is an **orthonormal basis** if

$$g(e_i, e_j) = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases}$$

where δ_{ij} , defined by the last equality, is called the **Kronecker delta**.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.12

A linear operator $\mathbf{T}$ of an inner product space $(\mathcal{V}, g)$ is $\mathbf{0}$ if and only if $g(a, \mathbf{T}b) = 0$ for all $a, b \in \mathcal{V}$.

Lemma 1.13

For any linear operator $\mathbf{T}$ and any linear combination $\alpha a + \beta b$ of $u, v \in \mathcal{V}$, one has the so-called **polarisation identity**

$$\begin{aligned} \alpha^* \beta g(a, \mathbf{T}b) + \alpha \beta^* g(b, \mathbf{T}a) = & g(\alpha a + \beta b, \mathbf{T}(\alpha a + \beta b)) \\ & - |\alpha|^2 g(a, \mathbf{T}a) - |\beta|^2 g(b, \mathbf{T}b). \end{aligned}$$

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.14

A linear operator $\mathbf{T}$ on an inner product space $(\mathcal{V}, g)$ is $\mathbf{0}$ if and only if $g(a, \mathbf{T}a) = 0$ for all $a \in \mathcal{V}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

*The set of vectors in $\mathcal{V}$ that are mapped onto the zero vector of $\mathcal{W}$ under the linear transformation $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{W}$ form a subspace of $\mathcal{V}$ called the **kernel** of $\mathbf{T}$ and denoted by $\ker \mathbf{T}$.*

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

*The range $\mathbf{T}(\mathcal{V})$ of a linear application $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{W}$ is a subspace of $\mathcal{W}$. The dimension of $\mathbf{T}(\mathcal{V})$ is called the **rank** of $\mathbf{T}$. The dimension of $\ker \mathbf{T}$ is called the **nullity** of $\mathbf{T}$.*

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.15

A linear transformation is injective if and only if its kernel consists only of the zero vector.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.16

A linear isometric map is injective.

Theorem 1.17 (Dimension Theorem)

Let $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{W}$ be a linear transformation. Then

$$\dim \mathcal{V} = \dim \ker \mathbf{T} + \dim \mathbf{T}(\mathcal{V}).$$

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Corollary 1.18

An endomorphism of a finite-dimensional linear space is bijective if and only if it is either injective or surjective.

Definition

A linear space $\mathcal{V}$ is said to be **isomorphic** to another linear space $\mathcal{W}$, and written $\mathcal{V} \cong \mathcal{W}$, if there exists a bijective linear application $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{W}$. In this case $\mathbf{T}$ is called a **linear isomorphism** (or simply, an **isomorphism**, when there is no risk of confusion). A bijective linear application of $\mathcal{V}$ onto itself is called an **automorphism** of $\mathcal{V}$. An automorphism is also called an **invertible map**. The set of automorphisms of $\mathcal{V}$ is denoted by $\text{GL}(\mathcal{V})$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Example 1.19

The set $\mathcal{M}_{\mathbb{C}}^{3 \times 3}$ of three by three matrices with complex entries is isomorphic to $\mathbb{C}^9$ as is the set of all polynomials of degree at most 8 with complex coefficients.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Corollary 1.20

An isometry of a finite-dimensional linear space is an automorphism of that linear space.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.21

A linear surjective map $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{W}$ is an isomorphism if and only if its nullity is zero.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.22

An injective linear transformation $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{W}$ carries linearly independent sets of vectors onto linearly independent sets of vectors.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.23

Two finite-dimensional vector spaces are isomorphic if and only if they have the same dimension.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Remark

A consequence of the previous theorem is that all N -dimensional vector spaces over $\mathbb{R}$ are isomorphic to $\mathbb{R}^N$ and all complex N -dimensional vector spaces are isomorphic to $\mathbb{C}^N$. So for all practical purposes, we have only two N dimensional vector spaces, $\mathbb{R}^N$ and $\mathbb{C}^N$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

A **linear functional** ϕ on a linear space $\mathcal{V}$ over a body $\mathbb{K}$ is a linear application $\phi \in \text{Hom}(\mathcal{V}, \mathbb{K})$, where the body of scalars $\mathbb{K}$ is regarded as a linear space.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

The **addition** and **multiplication by a scalar** of linear functionals are defined via

$$(\phi + \psi)(u) = \phi(u) + \psi(u) \quad \text{and} \quad (\alpha\phi)(u) = \alpha\phi(u),$$

for any $\phi, \psi \in \text{Hom}(\mathcal{V}, \mathbb{K})$, $u \in \mathcal{V}$ and $\alpha \in \mathbb{K}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Corollary 1.24

The set $\text{Hom}(\mathcal{V}, \mathbb{K})$ is a linear space over $\mathbb{K}$ with respect to the addition and multiplication by a scalar defined previously.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

*The linear space $\text{Hom}(\mathcal{V}, \mathbb{K})$ of all linear functionals on $\mathcal{V}$ is called the **dual space** of $\mathcal{V}$ and denoted by $\mathcal{V}'$.*

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Example 1.25

Let $(\mathcal{V}, g)$ be a complex inner product space. Fix $u \in \mathcal{V}$, and let $\gamma_u : \mathcal{V} \rightarrow \mathbb{C}$ be defined by

$$\gamma_u(v) = g(u, v).$$

Then γ_u is a linear functional.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.26

For every basis $B = \{e_j\}_{j=1}^N$ in a linear space $\mathcal{V}$, there corresponds a unique basis $B' = \{\phi_i\}_{i=1}^N$ in $\mathcal{V}'$ with the property that $\phi_i(e_j) = \delta_{ij}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

*The basis B' from the previous theorem is called the **dual basis** of B .*

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

Let $\{e_i\}_{i=1}^N$ be a basis of an inner product space $(\mathcal{V}, g)$, let $\mu_i = g(u, e_i)$ for some vector u . As noted earlier, the set of scalars $\{\mu_i\}_{i=1}^N$ defines a unique linear functional γ_u such that $\gamma_u(e_i) = \mu_i$. Then the vector γ_u of the dual space of $\mathcal{V}$ is called the **dual vector** of u .

Remark

The previous remarks show that for any vector u in a linear space $\mathcal{V}$ there corresponds a linear functional γ_u in the dual space $\mathcal{V}'$. The next notation was originally introduced by Paul Dirac, and in his original presentation, Dirac **assumed** a one-to-one correspondence between bras and kets, and it was not entirely clear whether this was a mathematical or physical assumption. Later on, when functional analysis was developed, the Riesz Representation Theorem showed there was no need, and indeed no room, for any such assumption, i.e., that there was always (in finite-dimensional spaces, and in Hilbert spaces) a one-to-one correspondence between vectors and linear functionals.

Dirac's (or "bracket") Notation

An arbitrary element of a complex linear space $\mathcal{V}$ is denoted $|\psi\rangle$ and is known as a **ket vector** (or simply, **ket**). The Dirac notation for the Hermitian conjugate (dual vector) of the ket vector $|\psi\rangle$ is written as $\langle\psi|$ and is known as a **bra vector** (or simply, **bra**). Given the two (contrived) names for the two sorts of vectors, an inner product like $\langle\psi|\phi\rangle$ becomes a particular type of "bra(c)ket". For operators acting on kets, we write $|\mathbf{T}a\rangle \equiv \mathbf{T}|a\rangle$. If $\mathcal{V} = \mathcal{L}$ and $\psi, \phi \in \mathcal{L}$, we define their **standard inner product**, denoted by $\langle\psi|\phi\rangle$, as

$$\langle\psi|\phi\rangle = \int_{\Omega} d\mu w \psi^* \phi$$

where Ω is a measurable set for which the integral —with respect to the measure μ — exists, and w , called the **weight function**, is a real-valued, continuous function that is always strictly positive on Ω . In particular, if $\psi, \phi \in \mathcal{L}(a, b)$, we usually take

$$\langle\psi|\phi\rangle = \int_{[a,b]} d\lambda w \psi^* \phi$$

with λ the Lebesgue measure on $\mathbb{R}$.



Paul Adrien Maurice
Dirac
1902-1984

Historical Notes

“Physical laws should have mathematical beauty”. This statement was Dirac’s response to the question of his philosophy of physics, posed to him in Moscow in 1955. He wrote it on a blackboard that is still preserved today.

Paul Adrien Maurice Dirac was born in 1902 in Bristol, England, of a Swiss, French-speaker father and an English mother. In the late 1920s the relentless march of ideas and discoveries had carried physics to a generally accepted relativistic theory of the electron. Dirac, however, was dissatisfied with the prevailing ideas and, somewhat in isolation, sought for a better formulation. By 1928 he succeeded in finding an equation, the *Dirac equation*, that accorded with his own ideas and also fit most of the established principles of the time. It was particularly remarkable for the internal beauty of its mathematical structure, which not only clarified previously mysterious phenomena such as spin and the **Fermi-Dirac** statistics associated with it, but also predicted the existence of an electron-like particle of negative energy, the antielectron, or positron, and, more recently, it has come to play a role of great importance in modern mathematics, particularly in the interrelations between topology, geometry, and analysis. **Heisenberg** characterised the discovery of antimatter by Dirac as “the most decisive discovery in connection with the properties or the nature of elementary particles.” One of the interesting implications of his work that predicted the positron was the prediction of a magnetic monopole. Dirac won the Nobel Prize in 1933 for this work.

“The amount of theoretical background one has to cover before being able to solve problems of real practical value is rather large, but this circumstance is an inevitable consequence of the fundamental part played by transformation theory and is likely to become more pronounced in the theoretical physics of the future”

– P.A.M. Dirac

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Example 1.27

Let $|e_k\rangle = e^{ikx}/\sqrt{2\pi}$ be functions in $\mathcal{L}(0, 2\pi)$ with $w(x) = 1$. Then show that $\langle e_l | e_k \rangle = \delta_{lk}$ where $\langle \cdot | \cdot \rangle$ is the standard inner product.

Dagger Notation

It is also convenient to introduce the **dagger operation** $\dagger : |a\rangle \mapsto \langle a|$ and use the notation

$$(|a\rangle)^\dagger = \langle a|,$$

where the symbol $\dagger$ means “dual, or dagger of”.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.28

The dagger operation acts on linear combinations as follows:

$$(\alpha |a\rangle + \beta |b\rangle)^\dagger = \alpha^* \langle a| + \beta^* \langle b|.$$

Definition

A **norm** on a linear space $\mathcal{V}$ is a functional $\|\cdot\| : \mathcal{V} \rightarrow \mathbb{R}$ satisfying the following properties:

- i) **(Positive definiteness)** $\|v\| \geq 0$ for all $v \in \mathcal{V}$ and $\|v\| = 0$ if and only if $v = 0$.
- ii) **(Homogeneity)** $\|\alpha v\| = |\alpha| \|v\|$ for any complex number α .
- iii) **(Triangle Inequality)** $\|u + v\| \leq \|u\| + \|v\|$.

A linear space on which a norm is defined is called a **normed linear space**.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.29 (Norms induce Metrics)

Any normed linear space induces a metric defined by

$$d(v, u) = \|v - u\| \quad \text{for all } v, u \in \mathcal{V}.$$

Theorem 1.30 (Cauchy-Bunyakovski-Schwarz Inequality)

For any pair of vectors u, v in an inner product space, the **Schwarz inequality** holds:

$$\langle u|u \rangle \langle v|v \rangle \geq |\langle u|v \rangle|^2.$$

Equality holds when u is proportional to v (when there exists a complex number α for which $u = \alpha v$).

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.31 (Inner Products induce Norms)

Any inner product space induces a norm defined by

$$||v|| = \sqrt{\langle v|v \rangle} \quad \text{for all } v \in \mathcal{V}.$$

Remark

The previous theorem shows that any inner product space is automatically a normed space, but the converse is not, in general, true. However, if the norm satisfies the **parallelogram law**,

$$\|a + b\|^2 + \|a - b\|^2 = 2\|a\|^2 + 2\|b\|^2,$$

then one can define

$$\langle a|b \rangle := \frac{1}{4} \{ \|a + b\|^2 - \|a - b\|^2 - i (\|a + ib\|^2 - \|a - ib\|^2) \}$$

and show that it is indeed an inner product. In fact, we have the following theorem.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.32

A normed linear space is an inner product space if and only if the norm satisfies the parallelogram law.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.33

Every finite-dimensional linear space can be turned into an inner product space.

Definition

Let $\mathcal{V}$ and $\mathcal{U}$ be inner product spaces. A homomorphism $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{U}$ is called an **isometric map** if

$$\langle \mathbf{T}a | \mathbf{T}b \rangle = \langle a | b \rangle ,$$

for all $|a\rangle, |b\rangle \in \mathcal{V}$. If $\mathcal{U} = \mathcal{V}$, then $\mathbf{T}$ is called an **isometry** of $\mathcal{V}$. An isometry of $\mathcal{V}$ is usually called a **unitary** (or **orthogonal**) operator.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Corollary 1.34

Isometries preserve (induced) norms.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

An **annihilator** of $|a\rangle \in \mathcal{V}$ is a linear functional $\phi \in \mathcal{V}'$ such that $\phi(|a\rangle) = 0$. Let $\mathcal{W}$ be a subspace of $\mathcal{V}$, then the set of linear functionals in $\mathcal{V}'$ that annihilate all vectors in $\mathcal{W}$ is denoted by $\mathcal{W}^0$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.35 (Dimension Theorem for Annihilators)

Let $\mathcal{W}$ be a subspace of $\mathcal{V}$. Then

$$\dim \mathcal{V} = \dim \mathcal{W} + \dim \mathcal{W}^0.$$

Definition

Let $\mathbf{T} : \mathcal{V} \rightarrow \mathcal{U}$ be a linear application. Define $\mathbf{T}^* : \mathcal{U}^* \rightarrow \mathcal{V}'$ by

$$[\mathbf{T}^*(\gamma)]|a\rangle = \gamma(\mathbf{T}|a\rangle) \quad \forall |a\rangle \in \mathcal{V}, \gamma \in \mathcal{V}'.$$

$\mathbf{T}^*$ is called the **dual** (or **pullback**) of $\mathbf{T}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.36

Let $\mathbf{T}$ be a linear homomorphism and let $\mathbf{T}^$ be its dual. Then $\ker \mathbf{T}^* = \mathbf{T}(\mathcal{V})^0$. If $\mathbf{T}$ is surjective [respectively, injective], then $\mathbf{T}^*$ is injective [respectively, surjective]. In particular, if $\mathbf{T}$ is an isomorphism, then so is $\mathbf{T}^*$.*

Definition

Let $\mathcal{V}$ and $\mathcal{U}$ be linear spaces. Let $\mathcal{V}^p$ denote the p -fold Cartesian product of $\mathcal{V}$. A **p -linear application** from $\mathcal{V}$ to $\mathcal{U}$ is a map $\theta : \mathcal{V}^p \rightarrow \mathcal{U}$ which is linear with respect to each of its arguments:

$$\begin{aligned} & \theta(|a_1\rangle, \dots, \alpha |a_j\rangle + \beta |b_j\rangle, \dots, |a_p\rangle) \\ &= \alpha \theta(|a_1\rangle, \dots, |a_j\rangle, \dots, |a_p\rangle) + \beta \theta(|a_1\rangle, \dots, |b_j\rangle, \dots, |a_p\rangle). \end{aligned}$$

The set of all p -linear applications from $\mathcal{V}$ to $\mathcal{U}$ is denoted by $\text{Hom}^p(\mathcal{V}, \mathcal{U})$. A p -linear application from $\mathcal{V}$ to $\mathbb{C}$ or $\mathbb{R}$ is called a **p -linear function** in $\mathcal{V}$.

Definition

Let σ denote a permutation of $1, 2, \dots, p$. Define the p -linear application $\sigma\omega$ by

$$\sigma\omega(|a_1\rangle, \dots, |a_p\rangle) = \omega(|a_{\sigma(1)}\rangle, \dots, |a_{\sigma(p)}\rangle).$$

An element $\omega \in \text{Hom}^p(\mathcal{V}, \mathcal{U})$ is **skew-symmetric** (or **alternating**) if

$$\sigma\omega = \epsilon_\sigma \cdot \omega, \text{ i.e., if } \omega(|a_{\sigma(1)}\rangle, \dots, |a_{\sigma(p)}\rangle) = \epsilon_\sigma \omega(|a_1\rangle, \dots, |a_p\rangle)$$

where ϵ_σ is the sign of σ . The set of p -linear skew-symmetric maps from $\mathcal{V}$ to $\mathcal{U}$ is denoted by $\Lambda^p(\mathcal{V}, \mathcal{U})$. A **p -form** is a p -linear alternating function in $\mathcal{V}$. The set of all p -forms in $\mathcal{V}$ is denoted by $\Lambda^p(\mathcal{V})$.

Proposition 1.37

Any p -linear application can be turned into a skew-symmetric p -linear application. In fact, if θ is a p -linear application, then

$$\omega \equiv \sum_{\pi \in S_p} \epsilon_{\pi} \cdot \pi \theta$$

is skew-symmetric.

Theorem 1.38

Let $\omega \in \text{Hom}^p(\mathcal{V}, \mathcal{U})$. Then the following statements are equivalent:

- i) $\omega(|a_1\rangle, \dots, |a_p\rangle) = 0$ whenever $|a_i\rangle = |a_j\rangle$ for some pair $i \neq j$.
- ii) $\omega(|a_{\sigma(1)}\rangle, \dots, |a_{\sigma(p)}\rangle) = \epsilon_\sigma \omega(|a_1\rangle, \dots, |a_p\rangle)$, for any permutation σ of $1, 2, \dots, p$, and any $|a_1\rangle, \dots, |a_p\rangle$ in $\mathcal{V}$.
- iii) $\omega(|a_1\rangle, \dots, |a_p\rangle) = 0$ whenever $\{|a_k\rangle\}_{k=1}^p$ are linearly dependent.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.39

Let $N = \dim \mathcal{V}$ and $\omega \in \Lambda^p(\mathcal{V}, \mathcal{U})$. Then ω is determined uniquely by its value on a basis of $\mathcal{V}$. In particular, if ω vanishes on a basis, then $\omega = \mathbf{0}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Definition

An N -form (element of $\Lambda^N(\mathcal{V})$) in an N -dimensional linear space $\mathcal{V}$ is called a **determinant function** in $\mathcal{V}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Proposition 1.40

In every finite-dimensional linear space, there are determinant functions which are not identically zero.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.41

Let $\omega \in \Lambda^N(\mathcal{V}, \mathcal{U})$. Let Δ be a fixed non-zero determinant function in $\mathcal{V}$. Then ω determines a unique $|u_\Delta\rangle \in \mathcal{U}$ such that

$$\omega(|v_1\rangle, \dots, |v_N\rangle) = \Delta(|v_1\rangle, \dots, |v_N\rangle) \cdot |u_\Delta\rangle.$$

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Corollary 1.42

Let Δ be a fixed nonzero determinant function in $\mathcal{V}$. Then every determinant function is a scalar multiple of Δ .

Definition

Let $\mathbf{A} \in \text{End}(\mathcal{V})$ and let $\{|e_i\rangle\}_{i=1}^N$ be a basis for $\mathcal{V}$. Let Δ be a nonzero determinant function in $\mathcal{V}$, and let Δ_A be defined by

$$\Delta_A(|e_1\rangle, \dots, |e_N\rangle) = \Delta(\mathbf{A}|e_1\rangle, \dots, \mathbf{A}|e_N\rangle).$$

Then

$$\Delta_A = \det \mathbf{A} \cdot \Delta$$

defines the **determinant** of $\mathbf{A}$.

Example 1.43

Find the determinants of the linear operators $\mathbf{T}, \mathbf{U} \in \text{End}(\mathbb{R}^2)$ defined by

$$\mathbf{T} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y \end{pmatrix} \quad \text{and} \quad \mathbf{U} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}, \quad a, b, c, d \in \mathbb{C}.$$

In particular, show that when $ad - bc = 0$, $\mathbf{U}$ is not invertible.

Theorem 1.44

The determinant of a linear operator $\mathbf{A}$ has the following properties:

- i) *If $\mathbf{A} = \lambda \mathbf{1}$, then $\det \mathbf{A} = \lambda^N$.*
- ii) *$\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$.*
- iii) *$\det(\mathbf{A} \circ \mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$.*

Theorem 1.45

Let Δ be a determinant function in the N -dimensional linear space $\mathcal{V}$. Let $|v\rangle$ and $\{|v_k\rangle\}_{k=1}^N$ be vectors in $\mathcal{V}$. Then

$$\sum_{j=1}^N (-1)^{j-1} \Delta(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_N\rangle) \cdot |v_j\rangle = \Delta(|v_1\rangle, \dots, |v_N\rangle) \cdot |v\rangle$$

where a hat on a vector means that particular vector is missing.

Definition

Let $\mathcal{V}$ be an N -dimensional linear space, Δ a determinant function in $\mathcal{V}$, and $\mathbf{A} \in \text{End}(\mathcal{V})$. For $|v\rangle, |v_i\rangle \in \mathcal{V}$, define $\Phi : \mathcal{V}^N \rightarrow \text{End}(\mathcal{V})$ by

$$\begin{aligned} & \Phi(|v_1\rangle, \dots, |v_N\rangle)|v\rangle \\ &= \sum_{j=1}^N (-1)^{j-1} \Delta(|v\rangle, \mathbf{A}|v_1\rangle, \dots, \widehat{\mathbf{A}|v_j\rangle}, \dots, \mathbf{A}|v_N\rangle) \cdot |v_j\rangle. \end{aligned}$$

Then the unique linear operator $\text{ad}(\mathbf{A})$ such that

$$\Phi(|v_1\rangle, \dots, |v_N\rangle) = \Delta(|v_1\rangle, \dots, |v_N\rangle) \cdot \text{ad}(\mathbf{A})$$

is called the **classical adjoint** of $\mathbf{A}$.

Groups

Bodies

Linear Spaces

Subspaces

Bases

Homomorphisms

Inner Products

Dimension Theorem

Isomorphisms

Dual Spaces

Dirac's Notation

Norms

Annihilators

Multilinear Maps

Determinants

Classical Adjoint

Theorem 1.46

The classical adjoint satisfies the following relations:

$$\text{ad}(\mathbf{A}) \circ \mathbf{A} = \det \mathbf{A} \cdot \mathbf{1} = \mathbf{A} \circ \text{ad}(\mathbf{A})$$

where $\mathbf{1}$ is the unit operator.

Corollary 1.47

A linear operator $\mathbf{A} \in \text{End}(\mathcal{V})$ is invertible if and only if $\det \mathbf{A} \neq 0$. In this case, we have

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \text{ad}(\mathbf{A}).$$